HomelessRegression

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Contents

Introduction

For this analysis, we are going to be using the following data from the DHS collected from homeless shelters in NYC from July 2015 to March 2025. At the time of writing this article, this dataset was last updated on August 3rd, 2025. source: https://catalog.data.gov/dataset/dhs-data-dashboard

This data set is about the total amount of homeless people held in shelters at the time of the census. This dataset breaks down the population by age, ethnicity, and gender. For the final columns, there are the "shelters placed"

There are a lot of features in this data set, here is a deconstruction of the different variable names:

- FWC -> Families with Children
- AF -> Adult Families
- SA -> Single Adults
- HoH -> Head of Households

NOTE TO READER

This data set changed halfway into 2022 its reporting frequency to be more frequent. To better explain the data, I decided to only keep the tail end of the data collected that month.

```r
paste("The amount of data points we have for this data set is: ", length(data[,1]))
```

```
## [1] "The amount of data points we have for this data set is:  121"
```

Regression

Regression Analysis

For this section, I will be running a regressive model for purely analysis. I will not be removing any datapoints as they are all statistically significant in building our understanding. As such, these models will have heavy skews. My goal in this section is to explore the data, and seeing what are the most influential features in finding shelter placements.

For this regression, we will be applying a Possion regression. A Poisson is most appropriate, as we want to look at the rate of acceptance of homeless people into shelters

Families with Children Regression Analysis

In this block, we apply a regression on all the datapoints that involve families with children. The analysis will be below the plots.

Part 0: Preparing the data to be modeled There are a couple of things we should do before handling the data, and that is to first isolate each individual group, and then create two seperate features for the month and year.

```
data_feat <- data %>%
  mutate(
    year  = year(Report.Date),
    month = factor(month(Report.Date))
  )
```

```
#regression based on shelter placements
library(dplyr)

fwc_reg <- data_feat %>%
  select(c("month", "FWC.Avg.Daily.census.Individual.in.Shelter...Adults",
           "FWC.Avg.Daily.census.Individual.in.Shelter...Children",
           "FWC.Unique.Individuals.by.Age...0.thru.5",
           "FWC.Unique.Individuals.by.Age...6.thru.13",
           "FWC.Unique.Individuals.by.Age...14.thru.17",
           "FWC..Unique.Individuals.by.Age..18.thru.20",
           "FWC.Unique.Individuals.by.Age...21.thru.29",
           "FWC.Unique.Individuals.by.Age...30.thru.44",
           "FWC.Unique.Individuals.by.Age...45.thru.64",
           "FWC.Unique.Individuals.by.Age...65.and.above",
           "FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander",
           "FWC.Race.Ethnicity.HoH...Black.non.Hispanic",
           "FWC.Race.Ethnicity.HoH...Hispanic",
           "FWC.Race.Ethnicity.HoH...Native.American",
           "FWC.Race.Ethnicity.HoH...White.non.Hispanic",
           "FWC.Race.Ethnicity.HoH...Unknown",
           "fwc_total_shelter_placements")  %>%
           na.omit())

target <- "fwc_total_shelter_placements"
```

```r
predictors <- setdiff(names(fwc_reg), target)

formula <- as.formula(
  paste(target, "~", paste(predictors, collapse = " + "))
)

model_fwc <- glm(formula, data = fwc_reg, family="poisson")
summary(model_fwc)
```

```
##
## Call:
## glm(formula = formula, family = "poisson", data = fwc_reg)
##
## Coefficients:
##                                                        Estimate Std. Error
## (Intercept)                                           4.886e+00  9.514e-02
## month2                                                1.023e-02  1.928e-02
## month3                                                1.166e-01  1.927e-02
## month4                                                1.031e-01  2.008e-02
## month5                                                8.980e-02  1.913e-02
## month6                                                1.115e-01  1.908e-02
## month7                                                3.903e-02  1.958e-02
## month8                                                4.365e-03  1.764e-02
## month9                                               -1.026e-01  1.769e-02
## month10                                              -6.302e-02  1.806e-02
## month11                                              -1.089e-01  1.869e-02
## month12                                               4.728e-02  1.750e-02
## FWC.Avg.Daily.census.Individual.in.Shelter...Adults   -3.709e-05  5.338e-06
## FWC.Avg.Daily.census.Individual.in.Shelter...Children -6.301e-05  5.669e-06
## FWC.Unique.Individuals.by.Age...0.thru.5             -7.342e-05  4.046e-05
## FWC.Unique.Individuals.by.Age...6.thru.13             6.483e-05  2.607e-05
## FWC.Unique.Individuals.by.Age...14.thru.17           -9.151e-05  1.134e-04
## FWC..Unique.Individuals.by.Age..18.thru.20            8.822e-04  1.398e-04
## FWC.Unique.Individuals.by.Age...21.thru.29            1.573e-05  3.237e-05
## FWC.Unique.Individuals.by.Age...30.thru.44            1.334e-04  5.159e-05
## FWC.Unique.Individuals.by.Age...45.thru.64           -4.710e-04  1.573e-04
## FWC.Unique.Individuals.by.Age...65.and.above          4.898e-03  7.009e-04
## FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander      -6.353e-05  6.587e-04
## FWC.Race.Ethnicity.HoH...Black.non.Hispanic           4.132e-04  3.374e-05
## FWC.Race.Ethnicity.HoH...Hispanic                     5.237e-05  3.319e-05
## FWC.Race.Ethnicity.HoH...Native.American             -7.335e-03  1.885e-03
## FWC.Race.Ethnicity.HoH...White.non.Hispanic          -2.371e-03  2.768e-04
## FWC.Race.Ethnicity.HoH...Unknown                      6.665e-04  8.633e-05
##                                                      z value Pr(>|z|)
## (Intercept)                                           51.361  < 2e-16 ***
## month2                                                 0.530 0.595924
## month3                                                 6.051 1.44e-09 ***
## month4                                                 5.133 2.86e-07 ***
## month5                                                 4.695 2.67e-06 ***
## month6                                                 5.843 5.12e-09 ***
## month7                                                 1.993 0.046234 *
## month8                                                 0.247 0.804558
## month9                                                -5.798 6.72e-09 ***
```

```
## month10                                                -3.490 0.000483 ***
## month11                                                -5.824 5.76e-09 ***
## month12                                                 2.702 0.006895 **
## FWC.Avg.Daily.census.Individual.in.Shelter...Adults    -6.949 3.67e-12 ***
## FWC.Avg.Daily.census.Individual.in.Shelter...Children -11.116  < 2e-16 ***
## FWC.Unique.Individuals.by.Age...0.thru.5               -1.815 0.069541 .
## FWC.Unique.Individuals.by.Age...6.thru.13               2.487 0.012900 *
## FWC.Unique.Individuals.by.Age...14.thru.17             -0.807 0.419774
## FWC..Unique.Individuals.by.Age..18.thru.20              6.311 2.76e-10 ***
## FWC.Unique.Individuals.by.Age...21.thru.29              0.486 0.627052
## FWC.Unique.Individuals.by.Age...30.thru.44              2.586 0.009714 **
## FWC.Unique.Individuals.by.Age...45.thru.64             -2.993 0.002760 **
## FWC.Unique.Individuals.by.Age...65.and.above            6.989 2.77e-12 ***
## FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander        -0.096 0.923169
## FWC.Race.Ethnicity.HoH...Black.non.Hispanic            12.247  < 2e-16 ***
## FWC.Race.Ethnicity.HoH...Hispanic                       1.578 0.114606
## FWC.Race.Ethnicity.HoH...Native.American               -3.891 1.00e-04 ***
## FWC.Race.Ethnicity.HoH...White.non.Hispanic            -8.564  < 2e-16 ***
## FWC.Race.Ethnicity.HoH...Unknown                        7.720 1.16e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 5728.43  on 120  degrees of freedom
## Residual deviance:  763.69  on  93  degrees of freedom
## AIC: 1825.6
##
## Number of Fisher Scoring iterations: 4
```

A couple of findings + A family is less likely to have a shelter placement based on whether they have children or not (WILL NEED CITATION) + Black and hispanic families are positively correlated with shelter placements

```
anova(model_fwc, test = "Chisq")
```

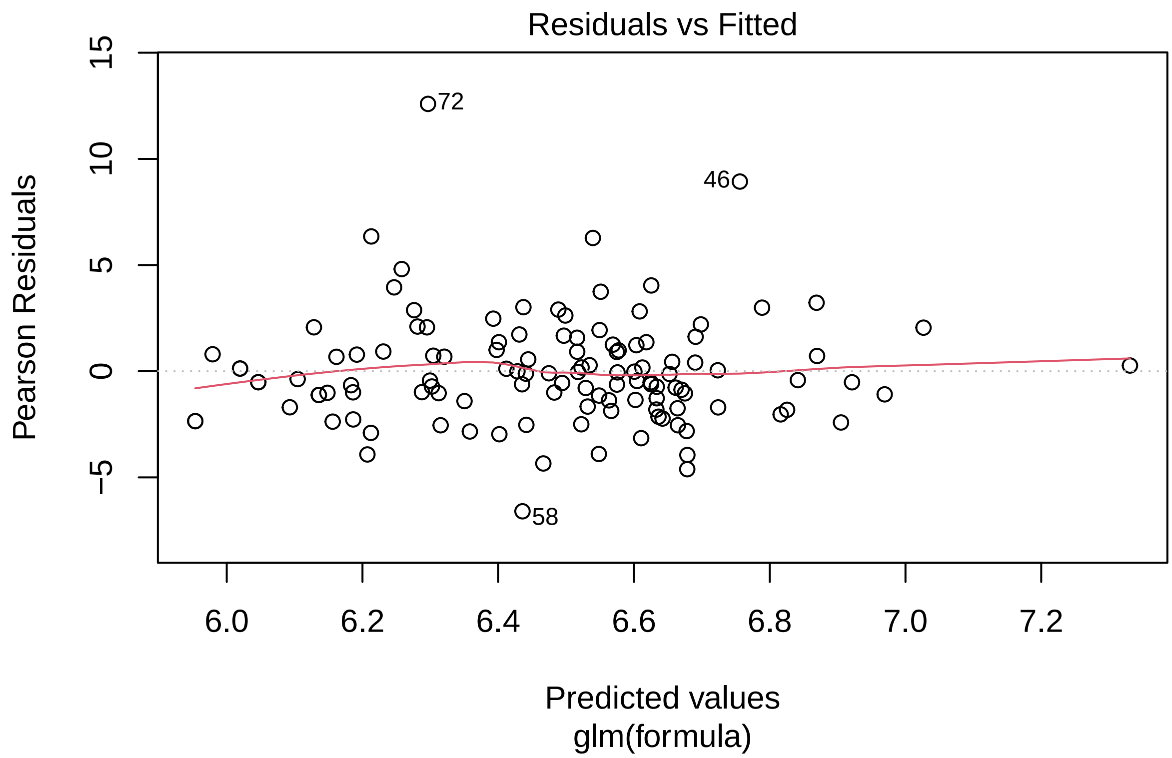
```
## Analysis of Deviance Table
##
## Model: poisson, link: log
##
## Response: fwc_total_shelter_placements
##
## Terms added sequentially (first to last)
##
##
##                                                       Df Deviance Resid. Df
## NULL                                                                    120
## month                                                 11   299.36       109
## FWC.Avg.Daily.census.Individual.in.Shelter...Adults    1  2095.32       108
## FWC.Avg.Daily.census.Individual.in.Shelter...Children  1   150.72       107
## FWC.Unique.Individuals.by.Age...0.thru.5               1   374.38       106
## FWC.Unique.Individuals.by.Age...6.thru.13              1   906.54       105
## FWC.Unique.Individuals.by.Age...14.thru.17             1   174.31       104
```

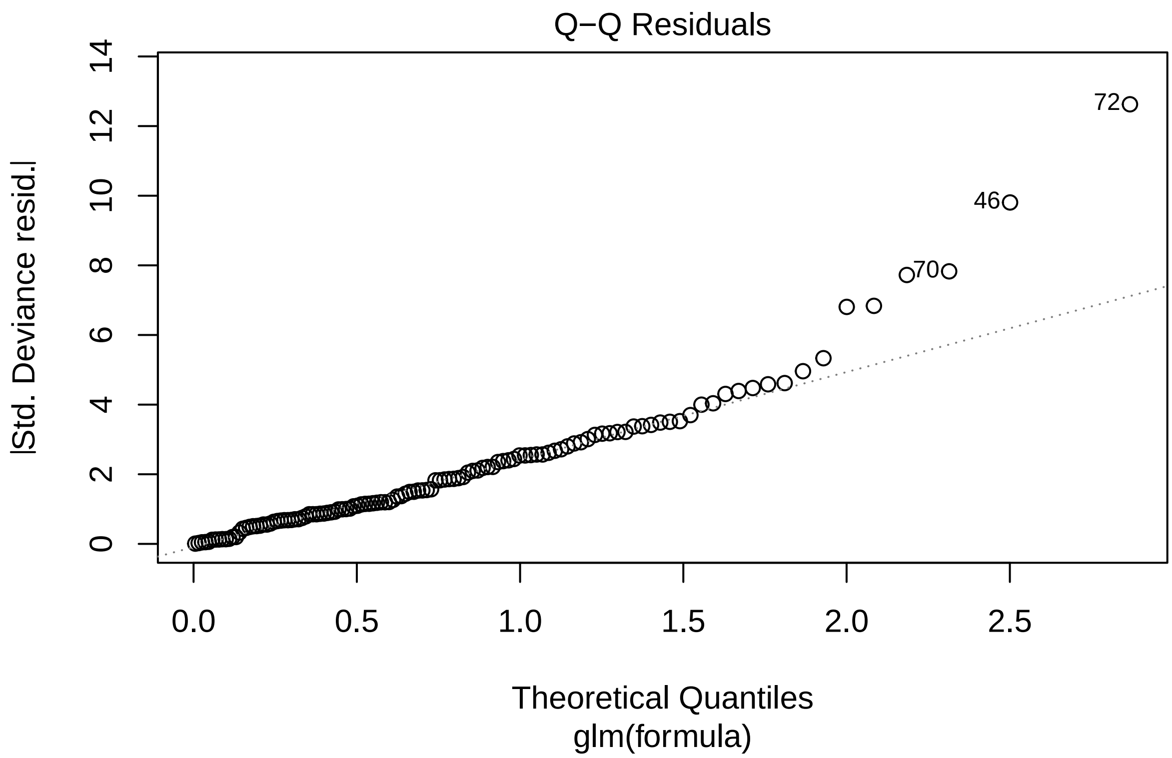
```
## FWC..Unique.Individuals.by.Age..18.thru.20            1   188.49       103
## FWC.Unique.Individuals.by.Age...21.thru.29            1   394.92       102
## FWC.Unique.Individuals.by.Age...30.thru.44            1     7.32       101
## FWC.Unique.Individuals.by.Age...45.thru.64            1    86.83       100
## FWC.Unique.Individuals.by.Age...65.and.above          1    15.25        99
## FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander       1    55.91        98
## FWC.Race.Ethnicity.HoH...Black.non.Hispanic           1   113.98        97
## FWC.Race.Ethnicity.HoH...Hispanic                     1     0.48        96
## FWC.Race.Ethnicity.HoH...Native.American              1    13.79        95
## FWC.Race.Ethnicity.HoH...White.non.Hispanic           1    26.95        94
## FWC.Race.Ethnicity.HoH...Unknown                      1    60.20        93
##                                                      Resid. Dev  Pr(>Chi)
## NULL                                                     5728.4
## month                                                    5429.1 < 2.2e-16 ***
## FWC.Avg.Daily.census.Individual.in.Shelter...Adults      3333.7 < 2.2e-16 ***
## FWC.Avg.Daily.census.Individual.in.Shelter...Children    3183.0 < 2.2e-16 ***
## FWC.Unique.Individuals.by.Age...0.thru.5                 2808.7 < 2.2e-16 ***
## FWC.Unique.Individuals.by.Age...6.thru.13                1902.1 < 2.2e-16 ***
## FWC.Unique.Individuals.by.Age...14.thru.17               1727.8 < 2.2e-16 ***
## FWC..Unique.Individuals.by.Age..18.thru.20               1539.3 < 2.2e-16 ***
## FWC.Unique.Individuals.by.Age...21.thru.29               1144.4 < 2.2e-16 ***
## FWC.Unique.Individuals.by.Age...30.thru.44               1137.1 0.0068285 **
## FWC.Unique.Individuals.by.Age...45.thru.64               1050.2 < 2.2e-16 ***
## FWC.Unique.Individuals.by.Age...65.and.above             1035.0 9.398e-05 ***
## FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander           979.1 7.600e-14 ***
## FWC.Race.Ethnicity.HoH...Black.non.Hispanic               865.1 < 2.2e-16 ***
## FWC.Race.Ethnicity.HoH...Hispanic                         864.6 0.4873164
## FWC.Race.Ethnicity.HoH...Native.American                  850.8 0.0002045 ***
## FWC.Race.Ethnicity.HoH...White.non.Hispanic               823.9 2.089e-07 ***
## FWC.Race.Ethnicity.HoH...Unknown                          763.7 8.574e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

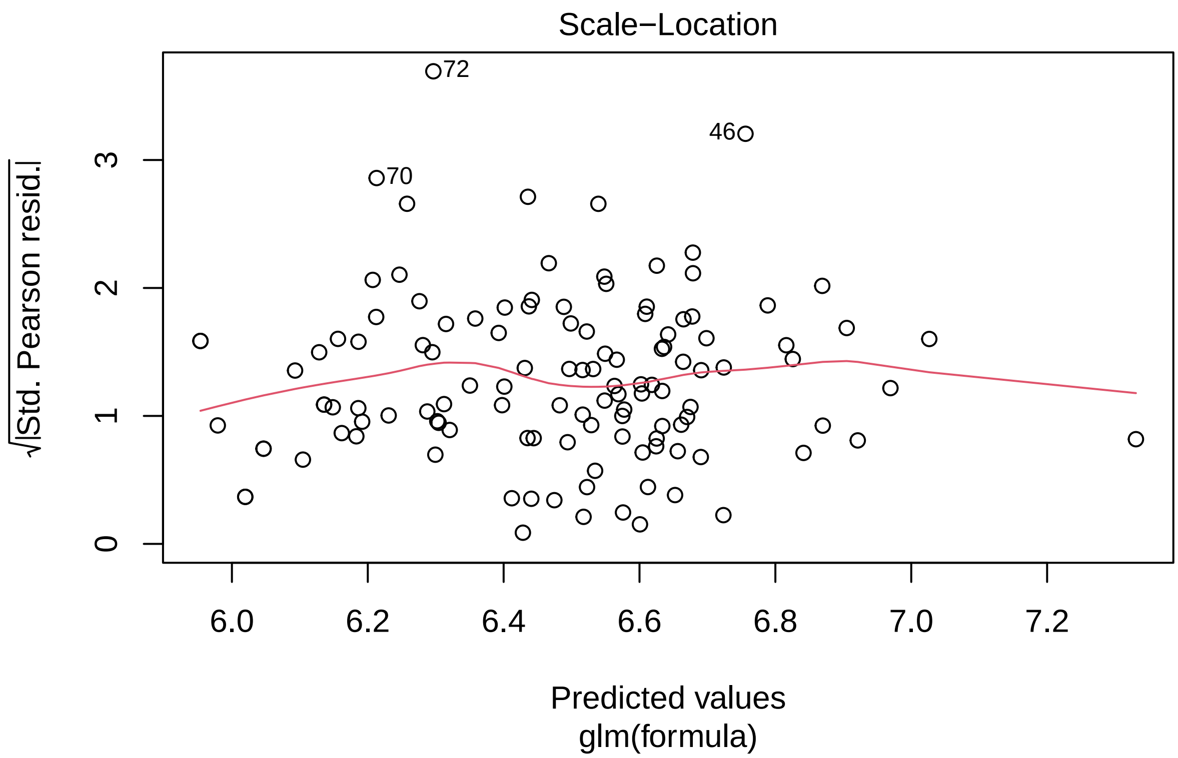
Couple of findings:

This model so far, will predict a negative value when

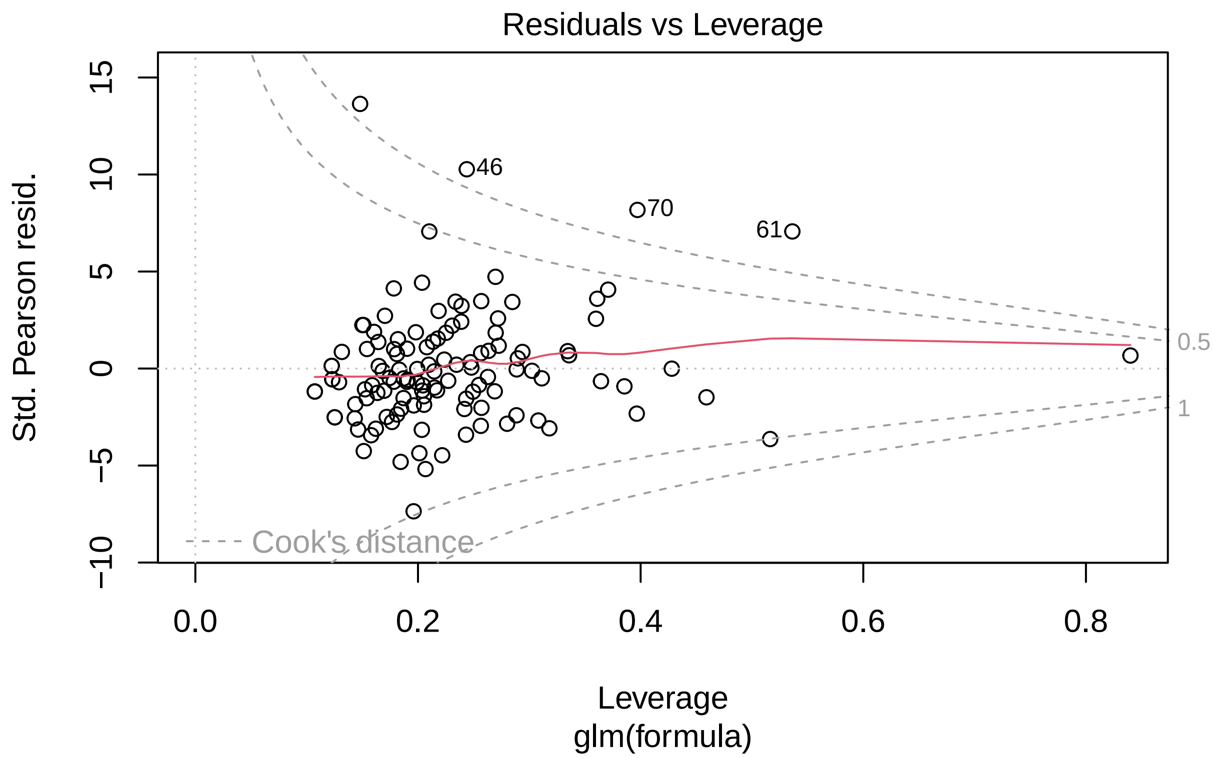
```r
plot(model_fwc)
```


Residuals vs Fitted
Pearson Residuals
15
10
5
0
−5
72
46
58
6.0
6.2
6.4
6.6
6.8
7.0
7.2
Predicted values
glm(formula)


Q−Q Residuals
|Std. Deviance resid.|
Theoretical Quantiles
glm(formula)
0
2
4
6
8
10
12
14
0.0
0.5
1.0
1.5
2.0
2.5
72
46
70



Scale−Location
√|Std. Pearson resid.|
0
1
2
3
72
46
70
6.0
6.2
6.4
6.6
6.8
7.0
7.2
Predicted values
glm(formula)




```
#There is a large tail but otherwise normal. Probably the surge in 2022
```

```
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode

## The following object is masked from 'package:purrr':
##
##     some
```

```
vif(model_fwc)
```

```
##                                                      GVIF Df
## month                                            4.376091 11
## FWC.Avg.Daily.census.Individual.in.Shelter...Adults     77.568531  1
## FWC.Avg.Daily.census.Individual.in.Shelter...Children   78.021588  1
```

```
## FWC.Unique.Individuals.by.Age...0.thru.5              796.072242  1
## FWC.Unique.Individuals.by.Age...6.thru.13             356.793367  1
## FWC.Unique.Individuals.by.Age...14.thru.17            969.241238  1
## FWC..Unique.Individuals.by.Age..18.thru.20            228.905990  1
## FWC.Unique.Individuals.by.Age...21.thru.29            349.578113  1
## FWC.Unique.Individuals.by.Age...30.thru.44           2878.832198  1
## FWC.Unique.Individuals.by.Age...45.thru.64            472.046152  1
## FWC.Unique.Individuals.by.Age...65.and.above           16.464526  1
## FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander        55.033574  1
## FWC.Race.Ethnicity.HoH...Black.non.Hispanic           106.643194  1
## FWC.Race.Ethnicity.HoH...Hispanic                     774.576741  1
## FWC.Race.Ethnicity.HoH...Native.American                4.760656  1
## FWC.Race.Ethnicity.HoH...White.non.Hispanic           117.781964  1
## FWC.Race.Ethnicity.HoH...Unknown                       60.255556  1
##                                                      GVIF^(1/(2*Df))
## month                                                       1.069400
## FWC.Avg.Daily.census.Individual.in.Shelter...Adults         8.807300
## FWC.Avg.Daily.census.Individual.in.Shelter...Children       8.832983
## FWC.Unique.Individuals.by.Age...0.thru.5                   28.214752
## FWC.Unique.Individuals.by.Age...6.thru.13                  18.888975
## FWC.Unique.Individuals.by.Age...14.thru.17                 31.132639
## FWC..Unique.Individuals.by.Age..18.thru.20                 15.129639
## FWC.Unique.Individuals.by.Age...21.thru.29                 18.697008
## FWC.Unique.Individuals.by.Age...30.thru.44                 53.654750
## FWC.Unique.Individuals.by.Age...45.thru.64                 21.726623
## FWC.Unique.Individuals.by.Age...65.and.above                4.057650
## FWC.Race.Ethnicity.HoH...Asian.Pacific.Islander             7.418462
## FWC.Race.Ethnicity.HoH...Black.non.Hispanic                10.326819
## FWC.Race.Ethnicity.HoH...Hispanic                          27.831219
## FWC.Race.Ethnicity.HoH...Native.American                    2.181893
## FWC.Race.Ethnicity.HoH...White.non.Hispanic                10.852740
## FWC.Race.Ethnicity.HoH...Unknown                            7.762445
```

```
#Insane multicollinearity, but how to handle this within the parameters of the question trying to be as
```